HDOC Day-59 Activity

Longest Substring Without Repeating Characters

Problem Statement

Write a C program that reads a string s and passes its address to a function named longest_unique_substring() using pointers. The function must find the longest substring containing no repeated characters, calculate and display its length, treat uppercase and lowercase letters as different characters, and display 0 if the input string is empty.

Algorithm

1. Start.

2. Declare a character array to store the input string.

3. Read the string, including spaces if present.

4. Pass the address of the string to longest_unique_substring().

5. Initialize an array to store the last-seen position of each character.

6. Traverse the string using a pointer; move the start of the current substring past a repeated character when needed.

7. Update the maximum substring length after each character.

8. Treat uppercase and lowercase as distinct characters.

9. If the input string is empty, the maximum length remains 0.

10. Display the maximum length and stop.

Test-Case Table

Test	Input String	Expected Length	Type
1	abcabcbb	3	Valid
2	bbbbb	1	Repeated characters
3	pwwkew	3	Valid
4	(empty)	0	Empty input
5	aAbBcC	6	Case-sensitive
6	abba	2	Overlapping repeats
7	a	1	Single character

C Program

```
#include <stdio.h>
#include <string.h>

void longest_unique_substring(char *s)
{
    int last[256], start = 0, maxLength = 0;
    int i, length;
    for (i = 0; i < 256; i++)
        last[i] = -1;
    for (i = 0; *(s + i) != '\0'; i++)
    {
        unsigned char ch = (unsigned char)*(s + i);
        if (last[ch] >= start)
```

```
            start = last[ch] + 1;
        last[ch] = i;
        length = i - start + 1;
        if (length > maxLength)
            maxLength = length;
    }
    printf("Length of longest substring: %d\n", maxLength);
}

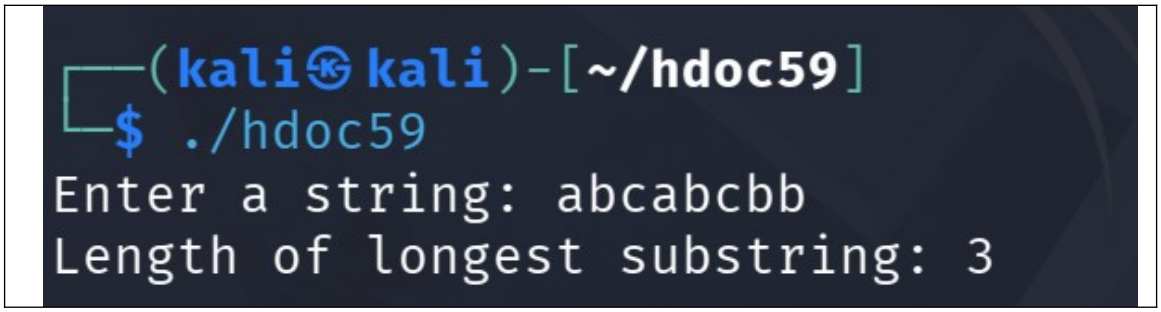
int main()
{
    char s[1000];
    printf("Enter a string: ");
    if (fgets(s, sizeof(s), stdin) == NULL)
        s[0] = '\0';
    s[strcspn(s, "\n")] = '\0';
    longest_unique_substring(s);
    return 0;
}
```

Compilation

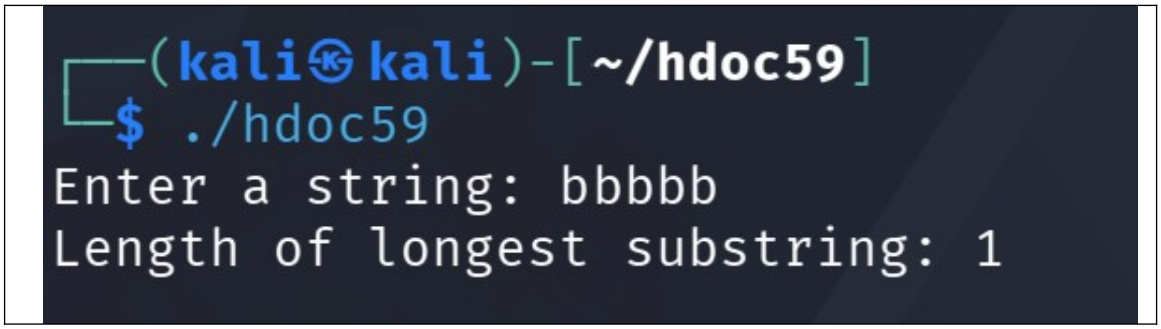


Execution

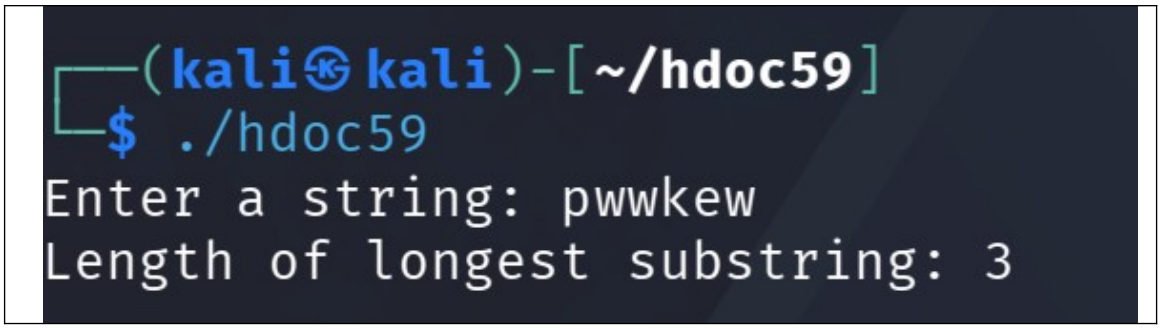
1. Main Execution - Example 1 (abcabcbb)



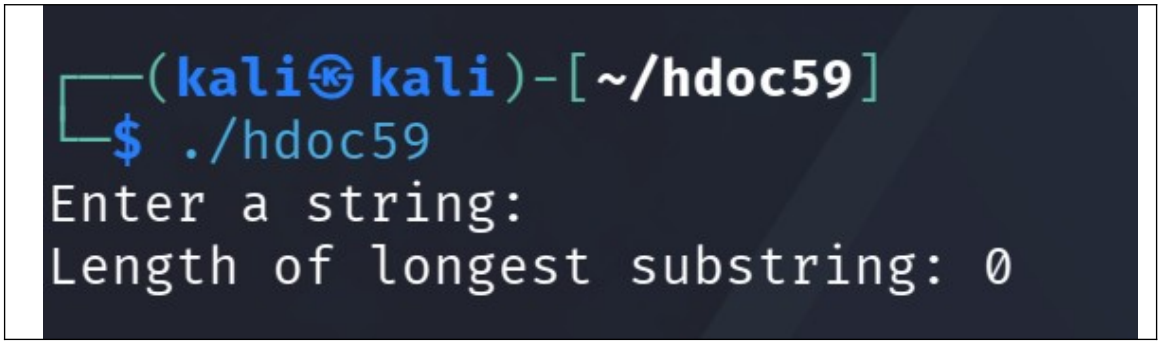
2. Example 2 (bbbbb)



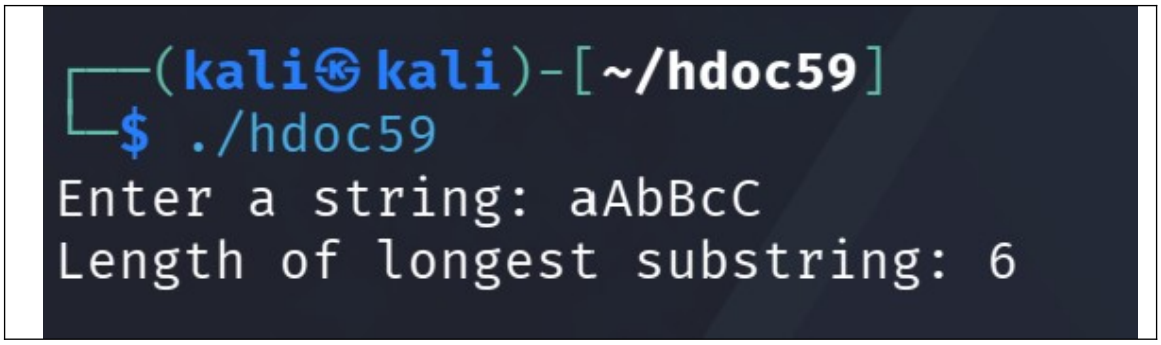
3. Example 3 (pwwkew)



4. Empty String



5. Case-Sensitive Input (aAbBcC)



Result

The C program uses pointers to pass the input string to longest_unique_substring() and calculates the length of the longest substring without repeating characters. It distinguishes uppercase and lowercase letters and returns 0 for an empty string.